



Dhirubhai Ambani University
Technology
Formerly DA-IICT

MediNexus Healthcare Ecosystem

Relational Schema, Minimal FD Set, BCNF Proofs, DDL Scripts

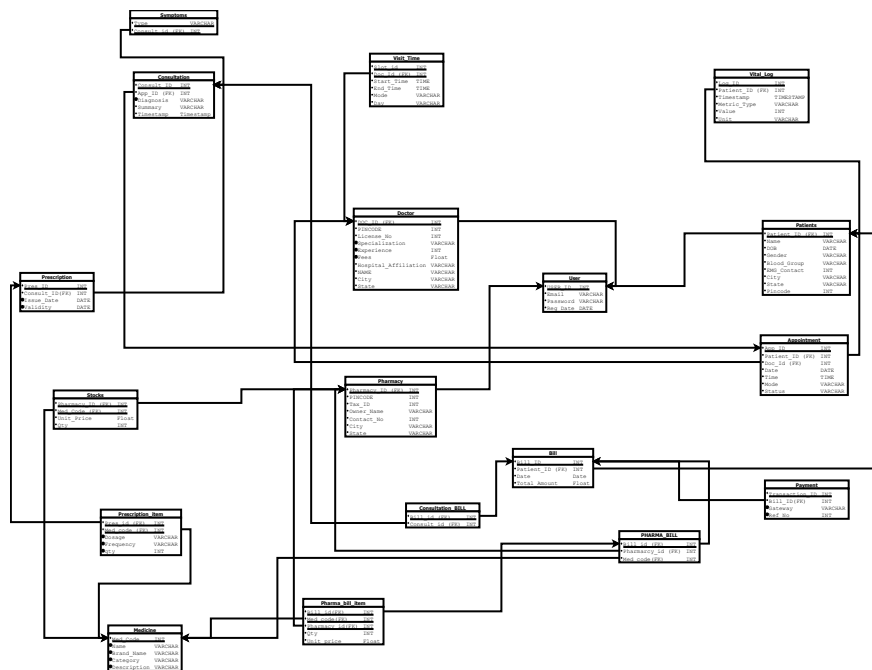
Course	IT214 - Database Management Systems
Semester	Winter 2026
Submission Date	4 April 2026

Project Team

Name	Roll Number
Atharv Trivedi	202401230
Jay Vaghela	202401237
Dharmraj Vaghela	202401236
Shlok Topaliya	202401229

Group Representative: Shlok Topaliya
Contact No.: +91 9082919527

Relational Schema



Functional Dependencies and BCNF Proof

Relational Schema Analysis

Note:

Some additional functional dependencies (such as uniqueness of Email, License_No, Tax_ID, Ref_No, and one-to-one mappings between Appointment–Consultation–Prescription and between Bill–Payment) are assumed as business constraints. Under these assumptions, all relations satisfy BCNF.

1. User

Candidate Keys: {USER_ID}, {Email}

Minimal FD Set:

- USER_ID → Email, Password, Reg_Date
- Email → USER_ID

BCNF Proof:

- USER_ID is a candidate key, so it is a superkey.
- Email is also a candidate key, so it is also a superkey.

Conclusion: User relation is in BCNF.

2. Patients

Candidate Key: {Patient_ID}

Minimal FD Set:

- Patient_ID → Name, DOB, Gender, Blood_Group, EMG_Contact, City, State, Pincode

BCNF Proof:

- Patient_ID is the only candidate key, so it is a superkey.
- No other non-trivial FD exists in this relation.

Conclusion: Patients relation is in BCNF.

3. Doctor

Candidate Keys: {DOC_ID}, {License_No}

Minimal FD Set:

- DOC_ID → License_No, Specialization, Experience, Fees, Hospital_Affiliation, Name, City, State, Pincode
- License_No → DOC_ID

BCNF Proof:

- DOC_ID is a candidate key.
- License_No is also unique for each doctor, so it is also a candidate key.

Conclusion: Doctor relation is in BCNF.

4. Medicine

Candidate Key: {Med_Code}

Minimal FD Set:

- Med_Code $\rightarrow$ Name, Brand_Name, Category, Description

BCNF Proof:

- Med_Code is the only candidate key.
- Hence, the determinant is a superkey.

Conclusion: Medicine relation is in BCNF.

5. Pharmacy

Candidate Keys: {Pharmacy_ID}, {Tax_ID}

Minimal FD Set:

- Pharmacy_ID $\rightarrow$ Tax_ID, Owner_Name, Contact_No, City, State, Pincode
- Tax_ID $\rightarrow$ Pharmacy_ID

BCNF Proof:

- Pharmacy_ID is a candidate key.
- Tax_ID is also unique, so it is also a candidate key.

Conclusion: Pharmacy relation is in BCNF.

6. Vital Log

Candidate Key: {Log_ID}

Minimal FD Set:

- Log_ID $\rightarrow$ Patient_ID, Timestamp, Metric_Type, Value, Unit

BCNF Proof:

- Log_ID uniquely identifies each record.

Conclusion: Vital Log relation is in BCNF.

7. Appointment

Candidate Key: {App_ID}

Minimal FD Set:

- App_ID $\rightarrow$ Patient_ID, Doc_ID, Date, Time, Mode, Status

BCNF Proof:

- App_ID is the primary identifier for each appointment.

Conclusion: Appointment relation is in BCNF.

8. Consultation

Candidate Keys: {Consult_ID}, {App_ID}

Minimal FD Set:

- Consult_ID $\rightarrow$ App_ID, Diagnosis, Summary, Timestamp
- App_ID $\rightarrow$ Consult_ID

BCNF Proof:

- Consult_ID is a candidate key.
- Each appointment leads to one consultation, so App_ID is also a candidate key.

Conclusion: Consultation relation is in BCNF.

9. Symptoms

Candidate Key: {Consult_ID, Type}

Minimal FD Set: No non-trivial functional dependency exists.

BCNF Proof:

- The only key is the full combination (Consult_ID, Type).
- Since no partial dependency exists, the relation is trivially in BCNF.

Conclusion: Symptoms relation is in BCNF.

10. Prescription

Candidate Keys: {Pres_ID}, {Consult_ID}

Minimal FD Set:

- Pres_ID $\rightarrow$ Consult_ID, Issue_Date, Validity
- Consult_ID $\rightarrow$ Pres_ID

BCNF Proof:

- Pres_ID is a candidate key.
- One consultation produces one prescription, so Consult_ID is also a candidate key.

Conclusion: Prescription relation is in BCNF.

11. Prescription Item

Candidate Key: {Pres_ID, Med_Code}

Minimal FD Set:

- (Pres_ID, Med_Code) → Dosage, Frequency, Qty

BCNF Proof:

- Both attributes together form the candidate key.
- Neither Pres_ID nor Med_Code alone can determine the other values.

Conclusion: Prescription Item relation is in BCNF.

12. Stocks

Candidate Key: {Pharmacy_ID, Med_Code}

Minimal FD Set:

- (Pharmacy_ID, Med_Code) → Unit_Price, Qty

BCNF Proof:

- The combination of pharmacy and medicine uniquely identifies stock details.

Conclusion: Stocks relation is in BCNF.

13. Visit Time (Slots)

Candidate Keys: {Slot_ID}, {Doc_ID, Day, Start_Time, Mode}

Minimal FD Set:

- Slot_ID → Doc_ID, Start_Time, End_Time, Mode, Day
- (Doc_ID, Day, Start_Time, Mode) → Slot_ID, End_Time

BCNF Proof:

- Slot_ID is a candidate key.
- The natural combination (Doc_ID, Day, Start_Time, Mode) is also unique.

Conclusion: Visit Time relation is in BCNF.

14. Bill

Candidate Key: {Bill_ID}

Minimal FD Set:

- Bill_ID $\rightarrow$ Patient_ID, Date, Total_Amount

BCNF Proof:

- Bill_ID is the only candidate key, so it is a superkey.
- Patient_ID does not determine Bill_ID (one patient may have many bills).
- No other non-trivial FD exists.

Conclusion: Bill relation is in BCNF.

15. Payment

Candidate Keys: {Transaction_ID}, {Ref_No}

Minimal FD Set:

- Transaction_ID $\rightarrow$ Bill_ID, Gateway, Ref_No
- Ref_No $\rightarrow$ Transaction_ID

BCNF Proof:

- Transaction_ID is a candidate key.
- Ref_No is also unique for each payment, so it is also a candidate key.

Conclusion: Payment relation is in BCNF.

16. Consultation Bill

Candidate Key: {Bill_ID, Consult_ID}

Minimal FD Set: No non-trivial functional dependency exists.

BCNF Proof:

- The relation contains only Bill_ID and Consult_ID; both together form the only candidate key.
- Since every attribute is part of the key, the relation is trivially in BCNF.

Conclusion: Consultation Bill relation is in BCNF.

17. PHARMA BILL

Candidate Key: {Bill_ID, Pharmacy_ID}

Minimal FD Set: No non-trivial functional dependency exists.

BCNF Proof:

- The relation contains only Bill_ID and Pharmacy_ID; both together form the only candidate key.
- Since every attribute is part of the key, the relation is trivially in BCNF.

Conclusion: PHARMA BILL relation is in BCNF.

18. Pharma Bill Item

Candidate Key: {Bill_ID, Med_Code, Pharmacy_ID}

Minimal FD Set:

- (Bill_ID, Med_Code, Pharmacy_ID) $\rightarrow$ Qty, Unit_Price

BCNF Proof:

- The full combination uniquely identifies each medicine entry in a pharmacy bill.

Conclusion: Pharma Bill Item relation is in BCNF.

All 18 relations satisfy BCNF under the stated business constraints.

DDL Script

Schema Initialisation

```

1 DROP SCHEMA IF EXISTS medinexus CASCADE;
2 CREATE SCHEMA medinexus;
3 SET search_path TO medinexus;

```

Listing 1: Drop and recreate the medinexus schema

Table Definitions & Sample Data

2.1 Table: users

```

1 CREATE TABLE medinexus.users (
2     user_id    SERIAL          PRIMARY KEY,
3     email      VARCHAR(100)    NOT NULL UNIQUE,
4     password   VARCHAR(255)    NOT NULL,
5     reg_date   DATE            NOT NULL DEFAULT CURRENT_DATE
6 );

```

Listing 2: DDL – users

```

1 INSERT INTO medinexus.users (email, password, reg_date) VALUES
2 ('arjun.mehta@gmail.com',      'hashed_pass_001', '2024-01-10'),
3 ('priya.sharma@gmail.com',    'hashed_pass_002', '2024-01-15'),
4 ('rohit.verma@gmail.com',     'hashed_pass_003', '2024-02-01'),
5 ('sneha.patel@gmail.com',     'hashed_pass_004', '2024-02-10'),
6 ('dr.kapoor@medinexus.com',   'hashed_pass_005', '2024-01-05'),
7 ('dr.nair@medinexus.com',     'hashed_pass_006', '2024-01-06'),
8 ('dr.desai@medinexus.com',    'hashed_pass_007', '2024-01-07'),
9 ('kavya.iyer@gmail.com',      'hashed_pass_008', '2024-03-01'),
10 ('mohit.jain@gmail.com',      'hashed_pass_009', '2024-03-05'),
11 ('dr.shah@medinexus.com',    'hashed_pass_010', '2024-01-08'),
12 ('medplus.ahmd@pharmacy.com', 'hashed_pass_011', '2024-01-09'),
13 ('apollo.gndh@pharmacy.com', 'hashed_pass_012', '2024-01-10'),
14 ('janaus.surat@pharmacy.com', 'hashed_pass_013', '2024-01-11'),
15 ('lifecare.vdra@pharmacy.com', 'hashed_pass_014', '2024-01-12');

```

Listing 3: Sample data – users (14 rows: 6 patients + 4 doctors + 4 pharmacies)

2.2 Table: patients

```

1 CREATE TABLE medinexus.patients (
2     patient_id INT          PRIMARY KEY
3                                     REFERENCES medinexus.users(user_id)
4                                     ON DELETE CASCADE,
5     name       VARCHAR(100) NOT NULL,
6     dob        DATE         NOT NULL,
7     gender     VARCHAR(10)  NOT NULL
8                                     CHECK (gender IN ('Male', 'Female', '
9                                     Other')),

```

```

9      blood_group VARCHAR(5),
10     emg_contact  VARCHAR(15),
11     city         VARCHAR(60),
12     state        VARCHAR(60),
13     pincode      CHAR(6)
14 );

```

Listing 4: DDL – patients

```

1 INSERT INTO medinexus.patients VALUES
2 (1, 'Arjun Mehta', '1995-06-15', 'Male', 'B+',
3  '9876543210', 'Ahmedabad', 'Gujarat', '380001'),
4 (2, 'Priya Sharma', '1998-11-22', 'Female', 'O+',
5  '9823456781', 'Gandhinagar', 'Gujarat', '382010'),
6 (3, 'Rohit Verma', '1990-03-08', 'Male', 'A+',
7  '9712345678', 'Surat', 'Gujarat', '395001'),
8 (4, 'Sneha Patel', '2000-07-19', 'Female', 'AB-',
9  '9654321098', 'Vadodara', 'Gujarat', '390001'),
10 (8, 'Kavya Iyer', '1993-09-30', 'Female', 'O-',
11  '9541234567', 'Anand', 'Gujarat', '388001'),
12 (9, 'Mohit Jain', '1988-12-05', 'Male', 'A-',
13  '9432109876', 'Rajkot', 'Gujarat', '360001');

```

Listing 5: Sample data – patients

2.3 Table: doctors

```

1 CREATE TABLE medinexus.doctors (
2     doc_id          INT          PRIMARY KEY
3                               REFERENCES medinexus.users
4                               (user_id)
5                               ON DELETE CASCADE,
6     name            VARCHAR(100) NOT NULL,
7     license_no      VARCHAR(30)  NOT NULL UNIQUE,
8     specialization  VARCHAR(100) NOT NULL,
9     experience      INT          CHECK (experience >= 0),
10    fees             NUMERIC(10,2) NOT NULL CHECK (fees >= 0)
11    ,
12    hospital_affiliation VARCHAR(150),
13    city             VARCHAR(60),
14    state            VARCHAR(60),
15    pincode          CHAR(6)
16 );

```

Listing 6: DDL – doctors

```

1 INSERT INTO medinexus.doctors VALUES
2 (5, 'Dr. Ramesh Kapoor', 'LIC-2001-GJ-001', 'Cardiologist',
3  20, 1500.00, 'Apollo Hospital Ahmedabad', 'Ahmedabad',
4  'Gujarat', '380015'),
5 (6, 'Dr. Meena Nair', 'LIC-2005-GJ-002', 'Dermatologist',
6  15, 1000.00, 'Sterling Hospital', 'Ahmedabad',

```

```

7      'Gujarat', '380052'),
8  (7, 'Dr. Suresh Desai', 'LIC-2008-GJ-003', 'Orthopedician',
9      12, 1200.00, 'Zydus Hospital', 'Ahmedabad',
10     'Gujarat', '380054'),
11  (10, 'Dr. Neha Shah', 'LIC-2010-GJ-004', 'General Physician',
12      8, 700.00, 'Civil Hospital Gandhinagar', 'Gandhinagar',
13     'Gujarat', '382010');

```

Listing 7: Sample data – doctors

2.4 Table: pharmacies

```

1 CREATE TABLE medinexus.pharmacies (
2     pharmacy_id    INT          PRIMARY KEY
3                     REFERENCES medinexus.users(
4                         user_id)
5                     ON DELETE CASCADE,
6     pharmacy_name  VARCHAR(150) NOT NULL,
7     tax_id         VARCHAR(30)  NOT NULL UNIQUE,
8     owner_name     VARCHAR(100) NOT NULL,
9     contact_no     VARCHAR(15)  NOT NULL,
10    city           VARCHAR(60),
11    state          VARCHAR(60),
12    pincode        CHAR(6)
13 );

```

Listing 8: DDL – pharmacies

```

1 INSERT INTO medinexus.pharmacies VALUES
2  (11, 'MedPlus Pharmacy Ahmedabad', 'GST-GJ-001-MED',
3     'Harish Patel', '07912345601', 'Ahmedabad', 'Gujarat', '380001'),
4  (12, 'Apollo Pharmacy Gandhinagar', 'GST-GJ-002-AP0',
5     'Sunita Mehta', '07912345602', 'Gandhinagar', 'Gujarat', '382010'),
6  (13, 'Jan Aushadhi Kendra Surat', 'GST-GJ-003-JAN',
7     'Rakesh Shah', '07912345603', 'Surat', 'Gujarat', '395001'),
8  (14, 'LifeCare Pharmacy Vadodara', 'GST-GJ-004-LIF',
9     'Anita Desai', '07912345604', 'Vadodara', 'Gujarat', '390001');

```

Listing 9: Sample data – pharmacies

2.5 Table: visit_time

```

1 CREATE TABLE medinexus.visit_time (
2     slot_id        SERIAL        PRIMARY KEY,
3     doc_id         INT           NOT NULL
4                     REFERENCES medinexus.doctors(doc_id)
5                     ON DELETE CASCADE,

```

```

6      day          VARCHAR(10) NOT NULL
7                                CHECK (day IN ('Monday','Tuesday','
8                                Wednesday',
9                                'Thursday','Friday','
10                               Saturday')),
11
12     start_time    TIME          NOT NULL,
13     end_time      TIME          NOT NULL,
14     mode          VARCHAR(20) NOT NULL
15                                CHECK (mode IN ('Online','Offline','
16                                Both')),
17     CONSTRAINT chk_slot_times CHECK (end_time > start_time)
18 );

```

Listing 10: DDL – visit_time

```

1  INSERT INTO medinexus.visit_time (doc_id, day, start_time,
2                                end_time, mode) VALUES
3  (5, 'Monday', '09:00', '13:00', 'Offline'),
4  (5, 'Wednesday', '09:00', '13:00', 'Offline'),
5  (5, 'Friday', '14:00', '18:00', 'Online'),
6  (6, 'Tuesday', '10:00', '14:00', 'Both'),
7  (6, 'Thursday', '10:00', '14:00', 'Both'),
8  (7, 'Monday', '14:00', '18:00', 'Offline'),
9  (7, 'Saturday', '09:00', '12:00', 'Offline'),
10 (10, 'Monday', '08:00', '12:00', 'Online'),
11 (10, 'Wednesday', '08:00', '12:00', 'Online'),
12 (10, 'Friday', '08:00', '12:00', 'Both');

```

Listing 11: Sample data – visit_time

2.6 Table: appointments

```

1  CREATE TABLE medinexus.appointments (
2      app_id          SERIAL          PRIMARY KEY,
3      patient_id     INT              NOT NULL
4                                REFERENCES medinexus.patients(
5                                patient_id)
6                                ON DELETE CASCADE,
7      doc_id         INT              NOT NULL
8                                REFERENCES medinexus.doctors(doc_id)
9                                ON DELETE CASCADE,
10     app_date        DATE              NOT NULL,
11     app_time        TIME              NOT NULL,
12     mode            VARCHAR(20) NOT NULL CHECK (mode IN ('Online','
13     Offline')),
14     status          VARCHAR(20) NOT NULL DEFAULT 'Scheduled'
15                                CHECK (status IN ('Scheduled','
16                                Completed',
17                                'Cancelled','No-
18                                Show'))
19 );

```

Listing 12: DDL – appointments

```

1 INSERT INTO medinexus.appointments
2   (patient_id, doc_id, app_date, app_time, mode, status)
3   VALUES
4   (1, 5, '2025-03-03', '10:00', 'Offline', 'Completed'),
5   (2, 6, '2025-03-04', '11:00', 'Offline', 'Completed'),
6   (3, 7, '2025-03-03', '15:00', 'Offline', 'Completed'),
7   (4, 10, '2025-03-05', '09:00', 'Online', 'Completed'),
8   (8, 5, '2025-03-05', '10:30', 'Offline', 'Completed'),
9   (9, 6, '2025-03-06', '10:00', 'Offline', 'Completed'),
10  (1, 10, '2025-04-02', '08:30', 'Online', 'Scheduled'),
11  (2, 7, '2025-04-05', '09:00', 'Offline', 'Scheduled'),
12  (3, 5, '2025-04-04', '14:00', 'Online', 'Cancelled'),
13  (4, 6, '2025-03-11', '11:00', 'Offline', 'No-Show');

```

Listing 13: Sample data – appointments

2.7 Table: consultations

```

1 CREATE TABLE medinexus.consultations (
2   consult_id SERIAL PRIMARY KEY,
3   app_id INT NOT NULL UNIQUE
4             REFERENCES medinexus.appointments(
5               app_id)
6             ON DELETE CASCADE,
7   diagnosis VARCHAR(500),
8   summary VARCHAR(1000),
9   timestamp TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
10 );

```

Listing 14: DDL – consultations

```

1 INSERT INTO medinexus.consultations (app_id, diagnosis, summary)
2   VALUES
3   (1, 'Hypertension Stage 1',
4     'Patient has elevated BP. Advised medication and diet control
5     .'),
6   (2, 'Eczema',
7     'Mild eczema on forearms. Prescribed topical cream.'),
8   (3, 'Hairline Fracture - Left Tibia',
9     'X-ray confirmed hairline fracture. Rest and support advised.
10    '),
11  (4, 'Viral Fever',
12    'Fever with body ache. Prescribed antipyretics and rest.'),
13  (5, 'Arrhythmia - Mild',
14    'Irregular heartbeat detected. ECG done. Medication
15    prescribed.'),
16  (6, 'Acne Vulgaris',
17    'Moderate acne. Prescribed antibiotics and face wash routine.
18    ');

```

Listing 15: Sample data – consultations

2.8 Table: symptoms

```

1 CREATE TABLE medinexus.symptoms (
2     consult_id INT NOT NULL
3     REFERENCES medinexus.consultations(
4         consult_id)
5     ON DELETE CASCADE,
6     type VARCHAR(100) NOT NULL,
7     PRIMARY KEY (consult_id, type)
8 );

```

Listing 16: DDL – symptoms

```

1 INSERT INTO medinexus.symptoms VALUES
2 (1, 'Headache'), (1, 'Dizziness'), (1, 'High Blood Pressure'),
3 (2, 'Itching'), (2, 'Redness'),
4 (3, 'Leg Pain'), (3, 'Swelling'),
5 (4, 'Fever'), (4, 'Body Ache'), (4, 'Fatigue'),
6 (5, 'Palpitations'), (5, 'Shortness of Breath'),
7 (6, 'Acne'), (6, 'Oily Skin');

```

Listing 17: Sample data – symptoms

2.9 Table: medicines

```

1 CREATE TABLE medinexus.medicines (
2     med_code SERIAL PRIMARY KEY,
3     name VARCHAR(150) NOT NULL,
4     brand_name VARCHAR(150),
5     category VARCHAR(80),
6     description VARCHAR(500)
7 );

```

Listing 18: DDL – medicines

```

1 INSERT INTO medinexus.medicines (name, brand_name, category,
2     description) VALUES
3 ('Amlodipine 5 mg', 'Amlokind', 'Antihypertensive',
4     'Calcium channel blocker'),
5 ('Betamethasone Cream', 'Betnovate', 'Topical Steroid',
6     'Reduces inflammation and itching'),
7 ('Ibuprofen 400 mg', 'Brufen', 'NSAID',
8     'Anti-inflammatory and pain reliever'),
9 ('Paracetamol 500 mg', 'Calpol', 'Antipyretic',
10    'Reduces fever and mild pain'),
11 ('Metoprolol 25 mg', 'Betaloc', 'Beta Blocker',
12    'Controls heart rate and BP'),
13 ('Doxycycline 100 mg', 'Doxt-SL', 'Antibiotic',

```

```

14      'Broad spectrum antibiotic for acne'),
15      ('Cetrizine 10 mg',      'Cetzine',      'Antihistamine',
16      'Relieves allergy symptoms'),
17      ('Pantoprazole 40 mg', 'Pan-40',      'PPI',
18      'Reduces stomach acid production'),
19      ('Vitamin D3 60000 IU','Uprise D3', 'Supplement',
20      'Weekly dose for Vitamin D deficiency'),
21      ('Azithromycin 500 mg','Azithral', 'Antibiotic',
22      'Used for bacterial infections');

```

Listing 19: Sample data – medicines

2.10 Table: prescriptions

```

1 CREATE TABLE medinexus.prescriptions (
2     pres_id      SERIAL PRIMARY KEY,
3     consult_id   INT      NOT NULL UNIQUE
4                 REFERENCES medinexus.consultations(
5                     consult_id)
6                 ON DELETE CASCADE,
7     issue_date   DATE      NOT NULL DEFAULT CURRENT_DATE,
8     validity     DATE      NOT NULL,
9     CONSTRAINT  chk_validity CHECK (validity >= issue_date)
10 );

```

Listing 20: DDL – prescriptions

```

1 INSERT INTO medinexus.prescriptions (consult_id, issue_date,
2                                     validity) VALUES
3 (1, '2025-03-03', '2025-06-03'),
4 (2, '2025-03-04', '2025-04-04'),
5 (3, '2025-03-03', '2025-05-03'),
6 (4, '2025-03-05', '2025-03-15'),
7 (5, '2025-03-05', '2025-06-05'),
8 (6, '2025-03-06', '2025-05-06');

```

Listing 21: Sample data – prescriptions

2.11 Table: prescription_items

```

1 CREATE TABLE medinexus.prescription_items (
2     pres_id      INT      NOT NULL
3                 REFERENCES medinexus.prescriptions(
4                     pres_id)
5                 ON DELETE CASCADE,
6     med_code     INT      NOT NULL
7                 REFERENCES medinexus.medicines(
8                     med_code)
9                 ON DELETE RESTRICT,
10    dosage        VARCHAR(60),
11    frequency     VARCHAR(60),
12    qty           INT      NOT NULL CHECK (qty > 0),

```

```

11     PRIMARY KEY (pres_id, med_code)
12 );

```

Listing 22: DDL – prescription_items

```

1 INSERT INTO medinexus.prescription_items VALUES
2 (1,1, '5 mg', 'Once daily at night', 30),
3 (1,8, '40 mg', 'Once daily before breakfast', 30),
4 (2,2, 'Apply thin layer', 'Twice daily', 1),
5 (2,7, '10 mg', 'Once daily at night', 10),
6 (3,3, '400 mg', 'Thrice daily after meals', 21),
7 (3,9, '60000 IU', 'Once a week', 4),
8 (4,4, '500 mg', 'Thrice daily', 15),
9 (4,10, '500 mg', 'Once daily for 3 days', 3),
10 (5,5, '25 mg', 'Once daily morning', 30),
11 (5,1, '5 mg', 'Once daily at night', 30),
12 (6,6, '100 mg', 'Once daily after meals', 30);

```

Listing 23: Sample data – prescription_items

2.12 Table: stocks

```

1 CREATE TABLE medinexus.stocks (
2     pharmacy_id INT NOT NULL
3     REFERENCES medinexus.pharmacies(
4         pharmacy_id)
5     ON DELETE CASCADE,
6     med_code INT NOT NULL
7     REFERENCES medinexus.medicines(
8         med_code)
9     ON DELETE RESTRICT,
10     unit_price NUMERIC(10,2) NOT NULL CHECK (unit_price >= 0),
11     qty INT NOT NULL CHECK (qty >= 0),
12     PRIMARY KEY (pharmacy_id, med_code)
13 );

```

Listing 24: DDL – stocks

```

1 INSERT INTO medinexus.stocks VALUES
2 (11,1, 45.00,200), (11,4, 12.00,500),
3 (11,5, 85.00,150), (11,8, 55.00,300),
4 (12,1, 47.00,180), (12,2,120.00, 80),
5 (12,6, 95.00,100), (12,7, 18.00,250),
6 (13,3, 25.00,400), (13,4, 11.00,600),
7 (13,9,210.00, 60), (13,10,75.00,120),
8 (14,2,115.00, 90), (14,5, 82.00,160),
9 (14,6, 92.00,110), (14,7, 17.50,280);

```

Listing 25: Sample data – stocks

2.13 Table: vital_logs

```

1 CREATE TABLE medinexus.vital_logs (
2     log_id          SERIAL          PRIMARY KEY,
3     patient_id     INT              NOT NULL
4                                     REFERENCES medinexus.patients(
5                                         patient_id)
6                                     ON DELETE CASCADE,
7     timestamp      TIMESTAMP        NOT NULL DEFAULT CURRENT_TIMESTAMP,
8     metric_type    VARCHAR(60)      NOT NULL,
9     value          INT              NOT NULL,
10    unit           VARCHAR(20)      NOT NULL
11 );

```

Listing 26: DDL – vital_logs

```

1 INSERT INTO medinexus.vital_logs
2     (patient_id, timestamp, metric_type, value, unit) VALUES
3 (1, '2025-03-03 09:30:00', 'Blood Pressure Systolic', 145, 'mmHg'),
4 (1, '2025-03-03 09:30:00', 'Blood Pressure Diastolic', 92, 'mmHg'),
5 (1, '2025-03-03 09:30:00', 'Heart Rate', 88, 'bpm'),
6 (2, '2025-03-04 10:15:00', 'Temperature', 99, 'F'),
7 (2, '2025-03-04 10:15:00', 'SpO2', 98, '%'),
8 (3, '2025-03-03 14:30:00', 'Blood Pressure Systolic', 118, 'mmHg'),
9 (3, '2025-03-03 14:30:00', 'Heart Rate', 76, 'bpm'),
10 (4, '2025-03-05 09:00:00', 'Temperature', 102, 'F'),
11 (4, '2025-03-05 09:00:00', 'Heart Rate', 96, 'bpm'),
12 (8, '2025-03-05 10:00:00', 'Blood Pressure Systolic', 130, 'mmHg'),
13 (8, '2025-03-05 10:00:00', 'Heart Rate', 82, 'bpm'),
14 (9, '2025-03-06 10:00:00', 'Temperature', 98, 'F'),
15 (9, '2025-03-06 10:00:00', 'SpO2', 97, '%');

```

Listing 27: Sample data – vital_logs

2.14 Table: bill

```

1 CREATE TABLE medinexus.bill (
2     bill_id        SERIAL          PRIMARY KEY,
3     patient_id     INT              NOT NULL
4                                     REFERENCES medinexus.patients(
5                                         patient_id)
6                                     ON DELETE RESTRICT,
7     bill_date      DATE              NOT NULL DEFAULT CURRENT_DATE,
8     total_amount   NUMERIC(10,2)    NOT NULL CHECK (total_amount >=
9     0)
10 );

```

Listing 28: DDL – bill

```

1 INSERT INTO medinexus.bill (patient_id, bill_date, total_amount)
2     VALUES
3 (1, '2025-03-03', 1500.00),
4 (2, '2025-03-04', 1000.00),

```

```

4 (3, '2025-03-03', 1200.00),
5 (4, '2025-03-05', 700.00),
6 (8, '2025-03-05', 1500.00),
7 (9, '2025-03-06', 1000.00),
8 (1, '2025-03-03', 850.00),
9 (2, '2025-03-04', 560.00),
10 (3, '2025-03-05', 320.00),
11 (4, '2025-03-06', 740.00);

```

Listing 29: Sample data – bill

2.15 Table: payments

```

1 CREATE TABLE medinexus.payments (
2     transaction_id SERIAL PRIMARY KEY,
3     bill_id INT NOT NULL UNIQUE
4     REFERENCES medinexus.bill(
5         bill_id)
6     ON DELETE RESTRICT,
7     gateway VARCHAR(60) NOT NULL,
8     ref_no VARCHAR(60) NOT NULL UNIQUE
9 );

```

Listing 30: DDL – payments

```

1 INSERT INTO medinexus.payments (bill_id, gateway, ref_no) VALUES
2 (1, 'Razorpay', 'REF-RZP-10001'),
3 (2, 'Paytm', 'REF-PTM-10002'),
4 (3, 'UPI', 'REF-UPI-10003'),
5 (4, 'NetBanking', 'REF-NB-10004'),
6 (5, 'Razorpay', 'REF-RZP-10005'),
7 (6, 'Paytm', 'REF-PTM-10006'),
8 (7, 'UPI', 'REF-UPI-10007'),
9 (8, 'Razorpay', 'REF-RZP-10008'),
10 (9, 'UPI', 'REF-UPI-10009'),
11 (10, 'Paytm', 'REF-PTM-10010');

```

Listing 31: Sample data – payments

2.16 Table: consultation_bills

```

1 CREATE TABLE medinexus.consultation_bills (
2     bill_id INT NOT NULL
3     REFERENCES medinexus.bill(bill_id)
4     ON DELETE CASCADE,
5     consult_id INT NOT NULL
6     REFERENCES medinexus.consultations(consult_id)
7     ON DELETE CASCADE,
8     PRIMARY KEY (bill_id, consult_id)
9 );

```

Listing 32: DDL – consultation_bills

```

1 INSERT INTO medinexus.consultation_bills VALUES
2 (1,1), (2,2), (3,3), (4,4), (5,5), (6,6);

```

Listing 33: Sample data – consultation_bills

2.17 Table: pharma_bills

```

1 CREATE TABLE medinexus.pharma_bills (
2     bill_id      INT NOT NULL
3                 REFERENCES medinexus.bill(bill_id)
4                 ON DELETE CASCADE,
5     pharmacy_id  INT NOT NULL
6                 REFERENCES medinexus.pharmacies(pharmacy_id)
7                 ON DELETE RESTRICT,
8     PRIMARY KEY (bill_id, pharmacy_id)
9 );

```

Listing 34: DDL – pharma_bills

```

1 INSERT INTO medinexus.pharma_bills VALUES
2 (7,11), (8,12), (9,13), (10,14);

```

Listing 35: Sample data – pharma_bills

2.18 Table: pharma_bill_items

```

1 CREATE TABLE medinexus.pharma_bill_items (
2     bill_id      INT NOT NULL
3                 REFERENCES medinexus.bill(bill_id)
4                 ON DELETE CASCADE,
5     med_code     INT NOT NULL
6                 REFERENCES medinexus.medicines(
7                     med_code)
8                 ON DELETE RESTRICT,
9     pharmacy_id  INT NOT NULL,
10    qty          INT NOT NULL CHECK (qty > 0),
11    unit_price    NUMERIC(10,2) NOT NULL CHECK (unit_price >= 0),
12    PRIMARY KEY (bill_id, med_code),
13    FOREIGN KEY (pharmacy_id, med_code)
14        REFERENCES medinexus.stocks(pharmacy_id, med_code)
15        ON DELETE RESTRICT
16 );

```

Listing 36: DDL – pharma_bill_items

```

1 INSERT INTO medinexus.pharma_bill_items VALUES
2 (7, 1, 11, 30, 45.00),
3 (7, 8, 11, 30, 55.00),
4 (8, 2, 12, 1, 120.00),
5 (8, 7, 12, 10, 18.00),
6 (9, 3, 13, 21, 25.00),

```

```
7 (9, 4, 13, 15, 11.00),  
8 (10, 5, 14, 30, 82.00),  
9 (10, 6, 14, 30, 92.00);
```

Listing 37: Sample data – pharma_bill_items

End of Script — 18 Tables